Economic Risk Analysis of a Copper Mining Project: A Monte Carlo Approach to Investment Valuation

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Abstract

This paper develops a stochastic valuation framework for a representative copper mining project under market and operational uncertainty. The model integrates a discounted cash flow structure with Monte Carlo simulation to jointly account for copper price dynamics, ore-grade evolution, and metallurgical recovery risk over a 15-year horizon. Copper prices are modeled through a Geometric Brownian Motion process calibrated with World Bank commodity data, while operational parameters are calibrated using mine-level and sector-level reports for Peru.

The simulation outputs indicate that expected project value remains positive, with an estimated Expected NPV of S/ 351,269,556. However, the downside tail is material: the probability of a negative NPV is 41.61%, the 5% Value at Risk is $-$S/ $1,311,282,054$, and the 5% Conditional Value at Risk reaches $-$S/ $1,599,912,510$. These results underscore the asymmetric risk profile of capital-intensive mining investments and support the use of probabilistic valuation tools in applied project appraisal.

1 Introduction

Large-scale mining projects are highly capital-intensive and structurally exposed to uncertainty. Investment performance depends not only on long-term commodity price paths, but also on operational factors such as ore quality deterioration and metallurgical efficiency. In practice, these sources of uncertainty interact and can substantially alter the economic viability of otherwise promising projects.

Standard deterministic discounted cash flow (DCF) models remain useful for baseline valuation, yet they are limited in their ability to represent distributional risk. Point-estimate assumptions can mask tail outcomes and understate downside exposure in sectors with high fixed costs and cyclical revenues.

This study proposes a Monte Carlo-based financial risk model for a copper mining investment. By explicitly modeling stochastic drivers and propagating them through annual cash flows, the framework estimates a full probability distribution of Net Present Value (NPV). Risk is summarized using metrics commonly used in applied finance and policy analysis, including the probability of loss, Value at Risk (VaR), and Conditional Value at Risk (CVaR).

Figure 1 summarizes the conceptual architecture linking stochastic inputs, cash flow generation, and risk measurement.

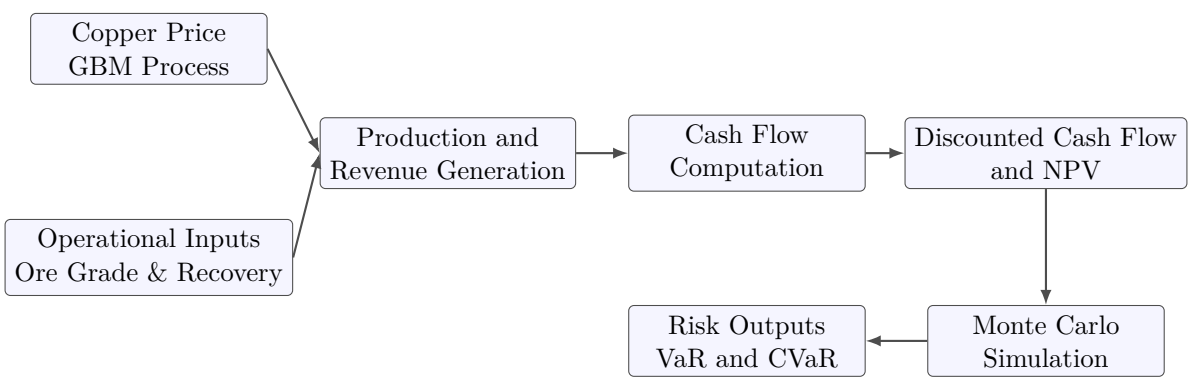

Figure 1: Conceptual Framework of the Economic Risk Model

2 Project Economic Framework

The project represents a hypothetical copper operation with a 15-year production horizon. Annual value is computed through a standard discounted cash flow structure:

$$NPV = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t}, \qquad T = 15, \tag{1}$$

where CF_t is net cash flow in year t and r is the discount rate.

To link physical production to financial outcomes, annual cash flow is defined as:

$$CF_t = \left(P_t \cdot Q_t\right) - OPEX_t - CAPEX_t - TAX_t, \tag{2}$$

with P_t denoting the copper price and Q_t denoting payable copper output. In turn, output depends on processed ore, ore grade, and metallurgical recovery:

$$Q_t = TON_t \cdot Grade_t \cdot Recovery_t. \tag{3}$$

Equations (1)–(3) establish the economic transmission channel through which market and operational shocks affect project valuation.

3 Commodity Price Modeling

Copper prices are modeled as a Geometric Brownian Motion (GBM), a standard process for assets with proportional shocks:

$$dP_t = \mu P_t \, dt + \sigma P_t \, dW_t, \tag{4}$$

where μ is the drift parameter, σ is instantaneous volatility, and W_t is a Wiener process. For simulation, the continuous-time representation is discretized as:

$$P_{t+1} = P_t \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_t\right], \tag{5}$$

with $\varepsilon_t \sim \mathcal{N}(0,1)$. Calibration based on historical series from the World Bank Commodity Price Data (Pink Sheet) yields representative parameters of $\mu \approx 0.05$ and $\sigma \approx 0.22$.

4 Operational Uncertainty

Operational risk is represented through ore grade and metallurgical recovery, both of which materially affect production and unit economics.

4.1 Ore Grade

Ore grade follows a declining structural trend associated with resource depletion, combined with stochastic deviations around that trend:

$$Grade_t = (\alpha + \beta t)\, \xi_t, \tag{6}$$

where $\beta < 0$ captures average grade deterioration and ξ_t is a truncated lognormal shock ensuring positive realizations.

4.2 Metallurgical Recovery

Metallurgical recovery measures the share of contained copper successfully extracted during processing. Recovery rates are modeled using a truncated normal distribution bounded to technically feasible values and calibrated with historical performance indicators for the Cerro Verde operation.

5 Data Sources

The model combines market and operational information from three primary sources:

- **World Bank Commodity Price Data (Pink Sheet):** monthly copper price series used to estimate GBM drift and volatility in Equations (4) and (5).
- **Freeport-McMoRan annual reports:** operational disclosures for Cerro Verde used to calibrate ore-grade and metallurgical recovery assumptions.
- **Peruvian Ministry of Energy and Mines reports:** sector production statistics used to benchmark and calibrate annual throughput and production levels.

This data combination supports an empirically grounded parameterization that is consistent with both international market dynamics and local production conditions.

6 Monte Carlo Simulation

The Monte Carlo engine generates 10,000 scenarios to propagate uncertainty from stochastic inputs to project valuation. In each scenario, copper prices follow Equation (5), while operational variables are jointly sampled from their calibrated distributions.

For each path, annual production, revenue, and cash flow are recomputed according to

Equations (2) and (3), and discounted to obtain NPV using Equation (1). The resulting empirical distribution of NPV forms the basis for risk quantification.

Relative to deterministic DCF analysis, this simulation-based approach retains information on distribution shape, downside tail mass, and dispersion, all of which are central for investment decisions under uncertainty.

7 Risk Metrics

Risk is summarized using distribution-based indicators derived from simulated NPV outcomes:

- Expected NPV, $\mathbb{E}[NPV]$.
- Probability of loss, $\Pr(NPV < 0)$.
- Value at Risk at 5%, defined as $VaR_{5\%} = F^{-1}_{NPV}(0.05)$.
- Conditional Value at Risk at 5%, defined as $CVaR_{5\%} = \mathbb{E}[NPV \mid NPV \leq VaR_{5\%}]$.

Together, these metrics characterize both central tendency and tail risk, improving comparability across alternative project designs or macroeconomic scenarios.

8 Results

The simulated NPV distribution is positively skewed but exhibits a heavy downside tail. Figure 2 shows the overall distribution, while Table 1 reports the core valuation and risk statistics.

Table 1: Summary Risk Indicators from Monte Carlo Simulation

Indicator	**Value**
Expected NPV	S/ 351,269,556
Probability of Loss, $\Pr(NPV < 0)$	41.61%
Value at Risk (5%)	$-$S/ $1,311,282,054$
Conditional Value at Risk (5%)	$-$S/ $1,599,912,510$

The expected NPV remains positive, suggesting that the project is attractive on average. However, the probability of loss above 40% indicates that mean outcomes alone are insufficient for decision-making. The joint behavior of $VaR_{5\%}$ and $CVaR_{5\%}$ further shows that adverse realizations are not only frequent enough to matter but also economically severe.

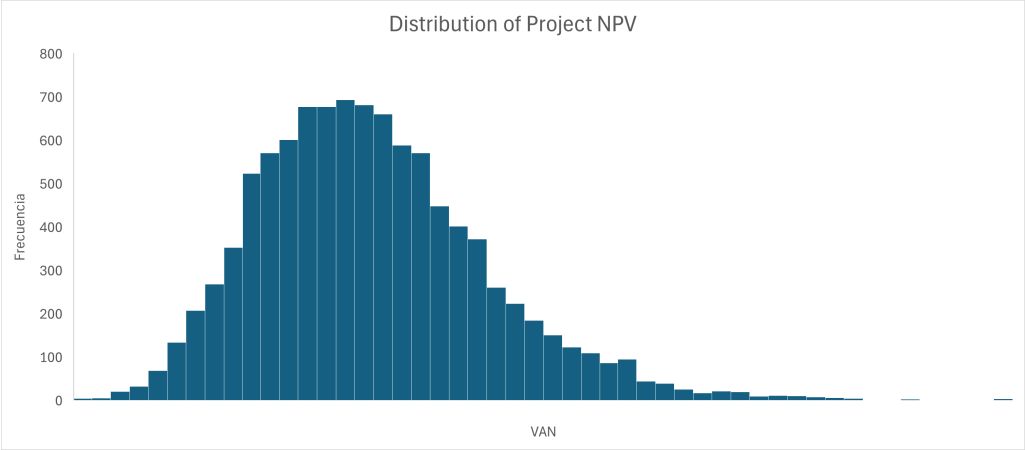

Figure 2: NPV Distribution from Monte Carlo Simulation

9 Economic Discussion

The results point to an asymmetric risk structure typical of extractive investments. Upside outcomes are driven by favorable price paths and stable operational performance, but downside scenarios can be disproportionately large because fixed costs and capital commitments amplify the effect of negative price shocks.

Copper price volatility is the dominant macro-financial driver of this asymmetry. Under GBM dynamics, periods of persistent low prices materially compress operating margins and magnify discounting effects on long-horizon cash flows. When these market shocks coincide with weaker ore grade or lower recovery, valuation losses become nonlinear.

Monte Carlo simulation improves on deterministic frameworks by preserving the full distribution of outcomes rather than a single-point estimate. This allows analysts and policymakers to evaluate trade-offs between expected value and tail exposure, and to incorporate risk-aware metrics such as VaR and CVaR into capital allocation, financing, and hedging decisions.

10 Conclusion

This paper presents a practical stochastic framework for evaluating mining investments under uncertainty. The model combines a DCF backbone with Monte Carlo simulation to integrate market risk and operational variability in a unified valuation exercise.

Although the expected NPV is positive, the estimated downside tail remains substantial, as reflected in the high loss probability and adverse CVaR outcomes. These findings reinforce the need for probabilistic appraisal methods in capital-intensive sectors exposed to commodity cycles.

Future extensions may incorporate endogenous operating flexibility, scenario-dependent financing costs, and policy sensitivity analysis to further strengthen project-level risk governance.